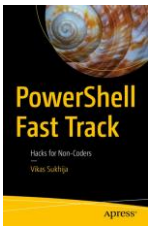
Week 1 Introduction to Cyber Security

This tutorial assumes you have already reviewed the lecture material.



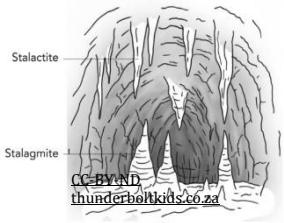
Readings

The following reading is available on the unit website:

- Chapter 6 of (Hassell 2017). Filtering and Limiting. This is an optional reading on filtering the pipeline using Where.

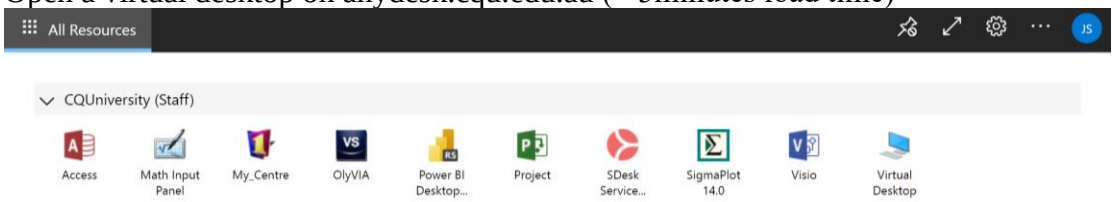
What is Cybersecurity?

Provide a stalactite and stalagmite analogy for cybersecurity threats, vulnerabilities, incidents, protections and risks.

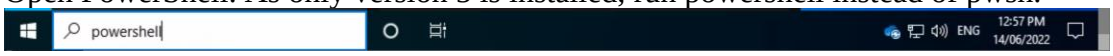


PowerShell

Open a virtual desktop on anydesk.cqu.edu.au (≈ 3minutes load time)



Open PowerShell. As only version 5 is installed, run powershell instead of pwsh.



Perform the following:

IP Addresses

Determine the IP address of the Ethernet 2 NIC.

```
PS> Get-Command "*IPAddress*"
```

Determine the network address of the first NIC.

Testing Connections

Use Test-NetConnection to ping www.cqu.edu.au to check if the host is up.

Test if the HTTP server at www.cqu.edu.au is working. First review the command options.

```
PS > man Test-NetConnection
```

Test if the HTTPS server at www.cqu.edu.au is working.

PowerShell Commands

Use the *ls* command to find the location of the *services* file somewhere in the *C:\windows\system32* folder. The *services* file lists default TCP and UDP port numbers. First, determine the options of *ls* that are used to recurse through all subdirectories, and to filter out strings.

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```
PS> help ls
```

Use the syntax 2>$null to redirect permission errors to the null file (bit bucket).

Use cat to review the services file to determine the TCP and UDP port numbers for the old game Doom by Id Software. Use the *sls* (Select-String) command to filter out those lines containing Doom.

Attribute Selection - Files
Use echo and > (output redirection) to create a file hello.txt and then use *ls* and the dot operator to output the LastWriteTime attribute of the hello.txt file.

Filter with Where-Object and Select Attributes with Select-Object
Finish the following command using Where-Object and Select-Object to output the LastWriteTime of the hello.txt file.
```
PS> ls | …
```

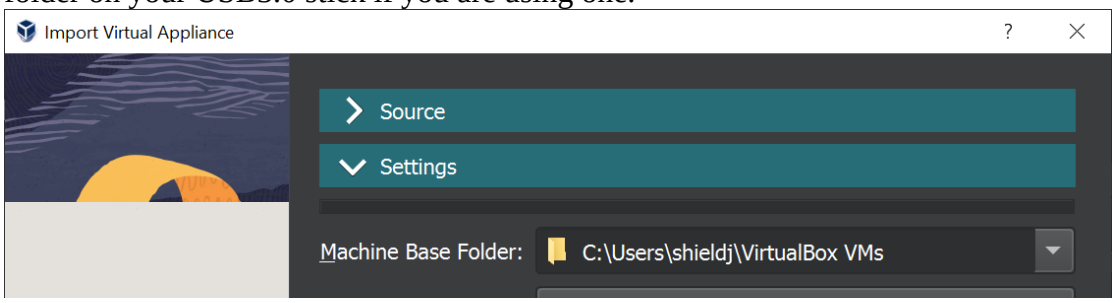
Explore Objects with Get-Member and Select multiple Attributes
Output the amount of free space on the drive on which the hello.txt file is stored. First, use *Get-Member* to determine the name of the drive property of the file.

Import the DC virtual machine into VirtualBox
Prerequisites
- Ensure you have at least 20Gb of free drive space to load a 6Gb virtual machine. Oncampus labs: you might need to use a 128Gb USB3.0 stick.
- You will need admin privileges if you do not already have Virtualbox installed on your computer.

Please:
- Install Virtualbox 7.
  - Oncampus labs: Virtualbox 6.1 should be installed and should work.
  - Choose one of the following installation options:
    - Chocolatey:
      - Install a Windows package manager such as chocolatey from https://chocolatey.org/
      - Start > PowerShell as Administrator.

      - The following will install the latest virtualbox:
        ```
        Admin Host PS> choco install virtualbox
        ```
        Do not type the > or anything before the > .
    - Winget:
      - Start PowerShell as Administrator as previously discussed.
      - ```
        PS> winget install --id=Oracle.VirtualBox  -e
        ```
        ```

        - Go to https://www.virtualbox.org/, download and install the latest Virtualbox.
- Download the DC .ova[1] image file from the unit website.
- Import the DC .ova file into VirtualBox: File > Import Appliance
    - Settings > Machine Base Folder: Set the Machine Base Folder to where you wish to import the VM. For example, you should set it to a folder on your USB3.0 stick if you are using one.

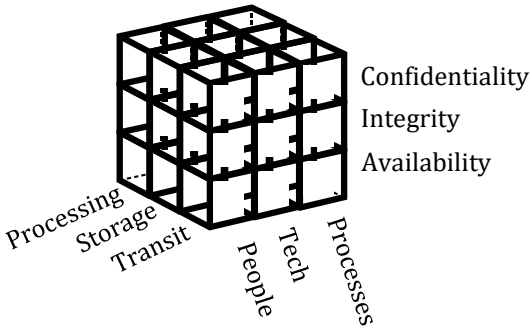


    - If you do not have sufficient disk space to import the virtual machine, you might receive an error such as Error 0x80070057.
- Delete the download image only if you need to free up disk space. Otherwise, keep the download image file so you can import the DC onto other hosts.

What are the three goals of cybersecurity?

What are the three features that define a data state and why do we consider data states?

McCumber's Cube
McCumber's cube can help identify cybersecurity considerations. Identify the McCumber planes for the following.



[1] Ova files are (uncompressed) tar files that contain an XML .ovf file that contains settings for, e.g. the network adapters and shared folders, virtual disks (.vmdk) and a .mf checksum file.

	Goal			Data State			Component/ Measure		
	Confidentiality	Integrity	Availability	Process	Storage	Transit	People	Tech	Process
Full disk encryption tool	☑	☐	☐	☐	☐	☐	☐	☐	☐
Organisation mandates employee checks prior to hiring	☐	☐	☐	☐	☐	☐	☐	☐	☐
Video demonstrating how to login using VPN	☐	☐	☐	☐	☐	☐	☐	☐	☐
Incident response manual for DDoS	☐	☐	☐	☐	☐	☐	☐	☐	☐

Use McCumber's cube to elicit some risks and measures for a patient database at a doctor's surgery. The database and web app are available on a local server which is accessed via web-browsers. Patients' details must be kept secret.

- Confidentiality/Storage/Tech: _____
- Confidentiality/Storage/Process: _____
- Confidentiality/Storage/People: _____

It is important the right patient is charged for services.

- Integrity/Processing/Process: _____

STRIDE-LM
What is STRIDE-LM?

STRIDE-LM Examples
The following are real Chromium bug reports. Classify their STRIDE-LM category.

- Issue 1245572: Google Chrome on Mac/Chromium can screen record even after the user unticks to disallow this action in MacOS' System Preferences, Security and Privacy, Privacy tab, Screen Recording.
- Issue 1269225: A specifically crafted HTML file can trigger a memory corruption in the renderer process. This bug may be potentially exploited to achieve one click remote code execution in the renderer process.
- Issue 727527: Password generation in Chrome (generate password option in chrome://flags) displays vulnerabilities.
    1. The final character in the password is always a digit
    2. The remaining characters are only chosen from [A-Za-z] - no special characters
    3. The password is of a fixed length (15 characters)

Incident Response Plans
What are some of the processes that need to be defined when planning responses to incidents such as a network intrusion or the loss of a mobile phone?

What are the learning outcomes of this unit?

What knowledge and skills are prerequisite for this unit?

List some of the solutions for securing computer networks and infrastructure (≈Week 8 of COIT11238).

List some of the aspects of a computer network that require management and/or some of the management tasks or tools (≈Week 11 of COIT11238).

References

Hassell, J 2017 *Learning PowerShell*. Boston, DeG Press. https://cqu.primo.exlibrisgroup.com/permalink/61CQU_INST/rpphdm/alma991001645923503441

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